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Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25EVT111 – ADVANCED BATTERY TECHNOLOGY FOR ELECTRIC VEHICLES

(For EVT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Identify the effects of temperature on VRLA and NiMH battery formation.	9	L*3	2	3
b) Explain the design of flooded lead acid battery.	7	L2	1	3
c) Illustrate the chemical functioning and reaction of fuel cell	4	L2	1	3
2. a) Examine the criteria that are useful to determine the end of formulation of a VRLA battery	9	L4	2	3
b) List the characteristics of the Li-ion battery that are favorable for EV applications	4	L2	1	3
c) Summarize are the salient features of the traction battery for EVs	7	L2	1	3
3. a) Enumerate the four steps of the conventional processing method	8	L2	1	3
b) Discuss the storage of VRLA and NiMH batteries	6	L3	2	3
c) Explain the effects of excessive heat on battery cycle life.	6	L3	2	3

Unit – II

4. a) Interpret the necessity of performing dynamic stress test (DST) for EV battery packs?	6	L3	4	3
b) List and analyze the observations made by the cold weather performance on a compact size EV using VRLA batteries?	6	L4	4	3
c) Explain the three primary methods of charging EV batteries.	8	L2	3	3
5. a) Utilize the initial conditions to successfully perform range tests on the EV.	9	L3	4	3
b) Explain conductive coupling and inductive coupling.	4	L2	3	3
c) Examine the AC impedance and AC conductance tests performed to determine battery pack impedance of the traction battery.	7	L2	3	3
6. a) Analyze the factors affecting the discharge of a traction battery.	9	L4	4	3
b) Outline the prerequisites for fast-charging?	7	L2	3	3
c) Explain the charging of NiMH batteries with the relevant chemical reactions that take place during charging process.	4	L2	3	3

Unit – III

7. a) Describe the BPMS charge protector and charge indicator	8	L2	5	3
b) Make use of the depolarization process to enhance the charging performance.	6	L3	5	3
c) Discover how a smart battery removes the charge control from the battery charger and assigns it to the battery.	6	L4	5	3
8. a) Summarize the key elements and functions of BPMS model.	7	L2	5	3
b) Explain the BPMS thermal management system.	7	L2	5	3
c) Analyze the key issues that need to be examined while selecting the types of fraction batteries.	6	L4	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
